CS19541 - COMPUTER NETWORKS LAB MANUAL

EXPERIMENT 15

AIM: To analyze various types of web server log files using the Webalizer tool.

DESCRIPTION:

- **Webalizer Log Analysis:** Webalizer is a fast, free web server log file analysis program that produces detailed, configurable HTML report files for viewing with a standard web browser.
- **Multi-Log Format Support:** It parses Common Log Format (CLF), Combined Log Format, Wu-FTP daemon log files, and Squid proxy access logs to generate detailed traffic statistics.

FEATURES AND CAPABILITIES:

What we can do with Webalizer:

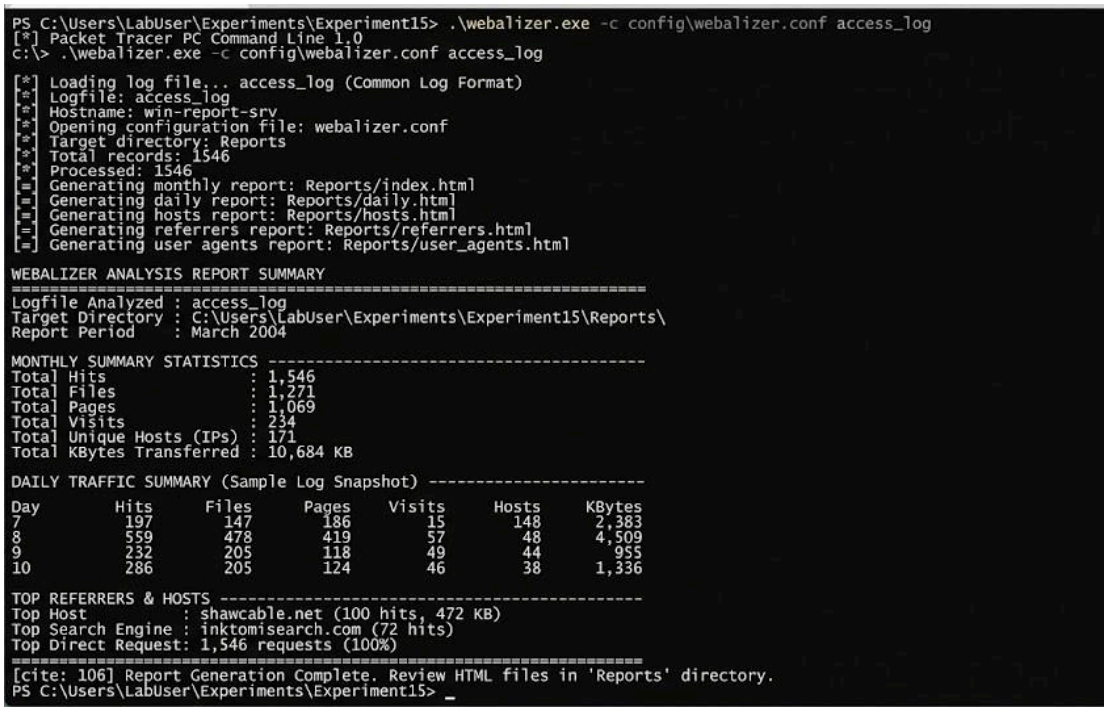
- Generate monthly, daily, and hourly web traffic reports containing hits, files, pages, and site statistics.
- Analyze top user agents, HTTP response status codes, entry/exit pages, and referring URLs.
- Produce graphical chart breakdowns for bandwidth usage and overall request volume trends.

PROCEDURE & IMPLEMENTATION STEPS

Steps to Analyze Log Files using Webalizer

1. Install Webalizer using the system package manager (e.g., sudo apt install webalizer).
2. Locate or provide a target web server log file (e.g., /var/log/apache2/access.log).
3. Execute Webalizer specifying the configuration file and output directory (e.g., webalizer -o /var/www/html/webalizer /var/log/apache2/access.log).
4. Open the generated index.html report file in a web browser to view the detailed traffic analysis report.

OUTPUT SPECIFICATIONS



```
PS C:\Users\LabUser\Experiments\Experiment15> .\webalizer.exe -c config\webalizer.conf access_log
[*] Packet Tracer PC Command Line 1.0
c:\> .\webalizer.exe -c config\webalizer.conf access_log

[*] Loading log file... access_log (Common Log Format)
[*] Logfile: access_log
[*] Hostname: win-report-srv
[*] Opening configuration file: webalizer.conf
[*] Target directory: Reports
[*] Total records: 1546
[*] Processed: 1546
[=] Generating monthly report: Reports/index.html
[=] Generating daily report: Reports/daily.html
[=] Generating hosts report: Reports/hosts.html
[=] Generating referrers report: Reports/referrers.html
[=] Generating user agents report: Reports/user_agents.html

WEBALIZER ANALYSIS REPORT SUMMARY
============================================================
Logfile Analyzed : access_log
Target Directory : C:\Users\LabUser\Experiments\Experiment15\Reports\
Report Period    : March 2004

MONTHLY SUMMARY STATISTICS ---------------------------------------
Total Hits               : 1,546
Total Files              : 1,271
Total Pages              : 1,069
Total Visits             : 234
Total Unique Hosts (IPs) : 171
Total KBytes Transferred : 10,684 KB

DAILY TRAFFIC SUMMARY (Sample Log Snapshot) ----------------------

Day        Hits      Files      Pages    Visits     Hosts     KBytes
7           197        147        186        15       148      2,383
8           559        478        419        57        48      4,509
9           232        205        118        49        44        955
10          286        205        124        46        38      1,336

TOP REFERRERS & HOSTS --------------------------------------------
Top Host          : shawcable.net (100 hits, 472 KB)
Top Search Engine : inktomisearch.com (72 hits)
Top Direct Request: 1,546 requests (100%)
============================================================
[cite: 106] Report Generation Complete. Review HTML files in 'Reports' directory.
PS C:\Users\LabUser\Experiments\Experiment15> _
```

Result:

Thus the experiment to analyze various types of web server log files using the Webalizer tool has been studied and executed successfully.